

MQST Ensemble QC Laboratory

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Lab 2 - Two Qubits

Winter quarter 2025

1 Introduction

In this second set of labs we will build on what we learned in Lab 1 and put in place the tools to be able to synthesize one- and two- qubit gates and make the leap to running gate-based quantum algorithms.

Real experiments on any real physical system require careful attention to the methods of data acquisition, an understanding of the contributions of noise, and techniques to optimize SNR (signal to noise ratio). Liquid state NMR is no different and you will first spend some time learning how to post-process the measured NMR signal rather than rely on the provided software to do this for you.

We will then begin to work with Chloroform, a prototypical “heteronuclear AX” system that will be used as our “two-qubit system” for most of the remaining labs. The main goal of this lab is to learn how to do pulsed control of Chloroform, which will involve doing a set of calibrations of the desired “single-qubit gates”, and of the essential ingredient of a “two-qubit gate” provided by the physical scalar (or J) coupling of the two nuclear spins in the Chloroform molecule.

I want to make a few comments about our molecular qubit system relative to the more ‘scalable’ qubit systems out there today. Not surprisingly, a system of “exchange coupled” nuclear spins looks a whole lot like that of semiconductor spin qubits. In fact the basic physical Hamiltonian is nearly identical, dominated by the combination of a Zeeman term and a scalar (or J) coupling.¹ The underlying physics are a bit different, but not that much, the main difference being the mechanism for the J coupling; for nuclear spins in a molecule this is an indirect spin-spin coupling mediated by the (very fast moving) electrons that covalently bond the molecule together, whereas for electron spins it is a result of the Heisenberg exchange interaction when the electron wave functions physically overlap one another.

¹G. Burkard, T. D. Ladd, J. M. Nichol, A. Pan, and J. R. Petta, Semiconductor Spin Qubits, ArXiv:2112.08863 [Cond-Mat, Physics:Physics, Physics:Quant-Ph] (2021).

Superconducting qubit systems (mostly transmon qubits these days)² and atomic qubits systems also require a scalar coupling term in order to do two-qubit gates. As there is no easily exploitable two-qubit physical interaction like in the case of spin (electron or nuclear) qubits, a little more ‘synthesis’ is required. With a transmon qubit for example, a common technique is to bring the two qubits into resonance with each other, turning on $\sigma_{\pm}$ terms in the Hamiltonian which together with additional phase gates or Hadamards allows the synthesis of a proper CNOT.³ For trapped ion qubits, it was the discovery of the Cirac-Zoller gate,⁴ that gave life to the field (don’t thrash me here Eric...), wherein an indirect coupling between pairs of ions is mediated by the collective motion of the whole (or part of the whole) ion system.

1.1 Assigned Reading

Please read (and re-read) the following sections in Keeler (first edition):

Acquisition and Signal Processing Chapters 4 and 5 (N.B. These are Chapters 5 and 13 of Keeler second edition)

Heteronuclear Two Spin Systems Section 2.4, Chapter 6 (especially sections 6.3-6.5)

Relaxation Phenomena Chapter 8 through page 8-10

1.2 Additional reading

The following are available in the Files section on BruinLearn and will be helpful in solidifying your understanding up to this point and to help prepare you for what’s to come :

- Mateescu and Valeriu, “2D NMR - Density Matrix and Product Operator Treatment”, file : MateescuValeriu-2dnmr-full.pdf
- MIT courseware : “Pulsed Nuclear Magnetic Resonance: Spin Echoes”, file : MIT-EchoNMRlab.pdf
- MIT courseware : “Quantum Information Processing with NMR”, file : MIT-QCwithNMRlab.pdf

²Superconducting flux qubits are inductively coupled to one another which does provide a direct scalar coupling. The issue is it is always on, hence the attraction of charge-based superconducting qubits that now dominate.

³P. Krantz, M. Kjaergaard, F. Yan, T. P. Orlando, S. Gustavsson, and W. D. Oliver, A Quantum Engineer’s Guide to Superconducting Qubits, Applied Physics Reviews 6, 021318 (2019).

⁴J. I. Cirac and P. Zoller, Quantum Computations with Cold Trapped Ions, Phys. Rev. Lett. 74, 4091 (1995).

2 Prelab

Do the assigned reading above and the following problems :

1. Consider the complex free induction decay signal $S(t) = e^{i\Omega t} e^{i\phi} e^{-t/T_2}$. Show analytically that the spectrum $R \cos(\phi) + I \sin(\phi)$ has a pure absorption lineshape. Plot this for various values of ϕ to see the effect of ‘phasing’ the signal.
2. For a time-domain signal in which the NMR signal has decayed long before the end of the acquisition time, explain why the SNR of the corresponding spectrum can be improved either by shortening the acquisition time, or by applying a suitable weighting function.
3. Keeler 1st edition : E 6-3
4. Despite what you have been told about the form of the J coupling term in the Hamiltonian, i.e. it is purely longitudinal $H_J = 2\pi J I_z S_z$, the full form of this term is

$$H_J = 2\pi J \hat{I} \cdot \hat{S} = 2\pi J (I_x S_x + I_y S_y + I_z S_z)$$

The full Hamiltonian of our two nuclear spin system is then

$$H_{IS} = \Omega_I I_z + \Omega_S S_z + H_J$$

Represent H_{IS} in usual four component basis ($|\alpha\alpha\rangle, |\alpha\beta\rangle, |\beta\alpha\rangle, |\beta\beta\rangle$). Assuming a heteronuclear system (different Ω 's), under what conditions is the approximation of a pure longitudinal coupling a good one, i.e. show which terms or parts of terms are dropped in this approximation and under what assumptions.

5. Show using the commutation relations of the spin operators that the four quantum coherence terms, $I_x S_x, I_x S_y, I_y S_x, I_y S_y$, do not evolve in the presence of a longitudinal coupling, $J I_z S_z$.
6. Terms like $I_x S_x$ would appear to represent states with measurable magnetization, for in this case both spins seem to have a net polarization in the $\hat{x}$ -direction (in the rotating frame). Show mathematically why there is no measurable magnetization in this (and the other similar states from the previous problem).
7. Consider a system composed of two spin- $\frac{1}{2}$ particles denoted by the spin operators I and S . A static magnetic field B_0 along the $\hat{z}$ -axis is applied to both spins, resulting in the Larmor frequencies ω_I and ω_S . Assume that the initial density matrix of the composite system is written as $\rho(0) = (\frac{1}{2}E + I_z) \otimes (\frac{1}{2}E + S_z)$.
 - (a) At $t = 0$ an ideal $\pi/2)_x$ pulse (delta function) is applied on spin I . Use QuTip to evaluate the density matrix after the rotation.

- (b) Immediately following the pulse, the two spins interact with the Hamiltonian $H_{IS} = 2\pi J(I_z \otimes S_z)$. Use QuTip to evaluate the density matrix $\rho(t)$.
- (c) Assuming that the observable operator is $O = I_x \otimes E + E \otimes S_x$, plot the observable signal for each of the spins using QuTip. [N.B. Don't forget that if you are observing just one of the spins, you just consider that term in the measurement operator!]

3 Lab Procedure

Read the material and follow the instructions in the Python notebook *NMR_Lab2.ipynb* found on BruinLearn and on the Magritek systems.